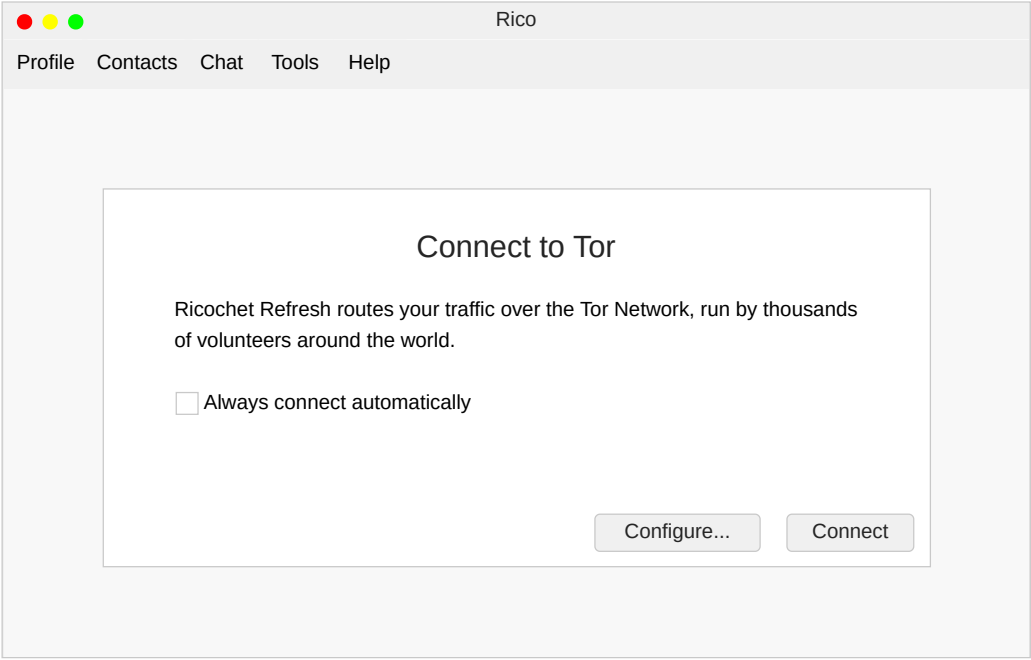


Rico - Connect

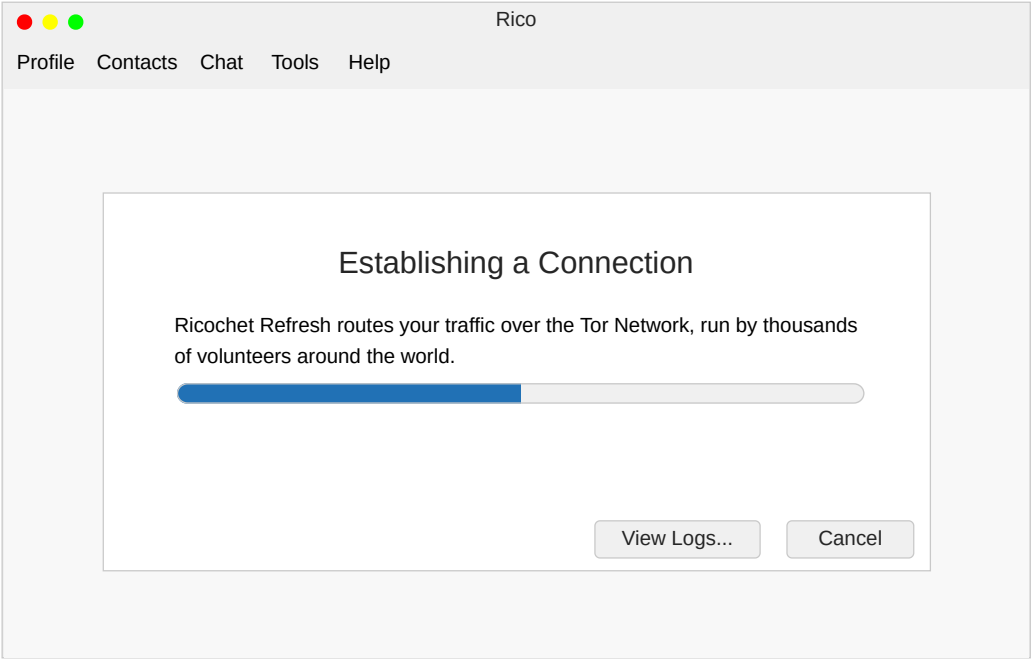
Users first reach this screen on application launch. At this point, the application is not yet connected to the Tor Network. This screen will allow the user to connect and/or configure the application preferences (including connection settings).



We will try to follow a desktop'ified version of Tor Browser's [about:torconnect](#) page here

Configure opens the Preferences dialog

Connect begins bootstrapping tor

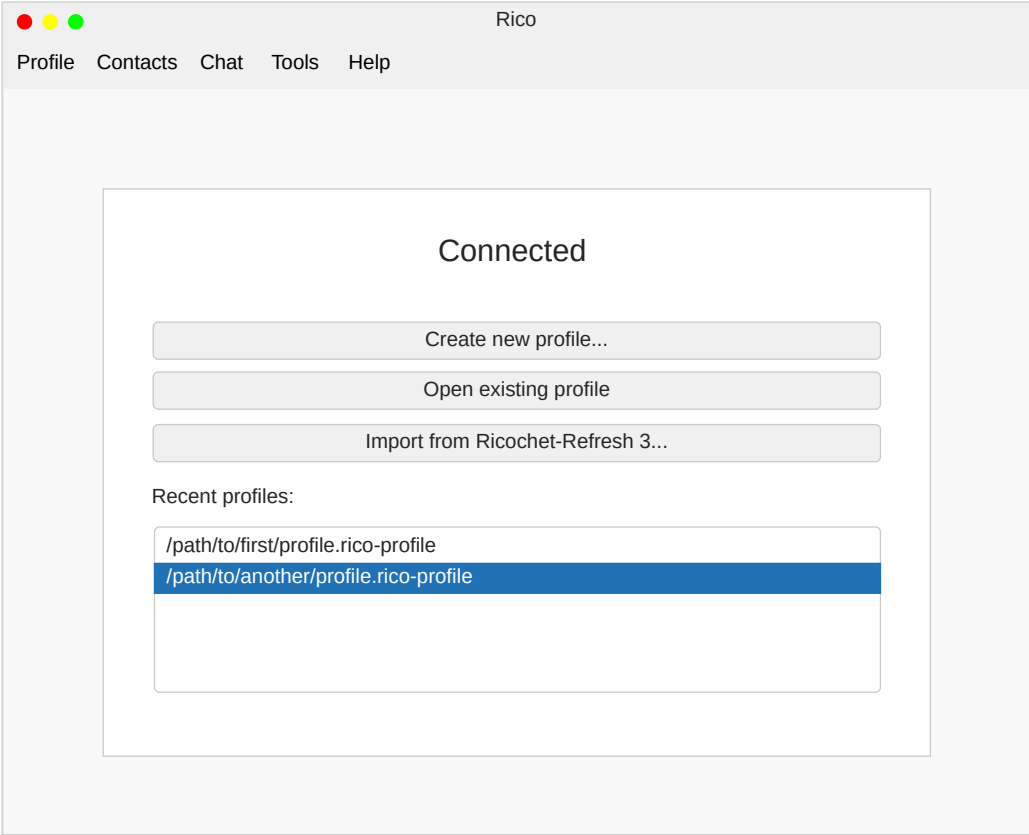


Open the Tor Logs dialog

Cancel current bootstrap attempt

Rico - Connected

User reaches this page after succesfully connecting to the Tor Network



Recent profile list can be disabled in application settings

Rico - Connection Status

Provide logs from the underlying Tor Provider. Some Tor Providers will not have logs to display (eg System Tor Daemon)

Other possible backends:

- system tor daemon
- in-proc arti-client
- bundled arti daemon
- system tor daemon

Statuses:

- Offline
- Connecting
- Connected

Rico

Profile

Contacts

Chat

Tools

Help

Connection Status

Tor backend: Bundled tor daemon (version 0.4.9.11)

Status: Offline

Aug 25 10:19:34.314 [notice] Tor 0.4.9.11 running on Linux with Libevent 2.1.12-stable, OpenSSL 3.0.20, Zlib 1.2.13, Liblzma 5.4.1, Libzstd 1.5.4 and Glibc 2.36 as libc.

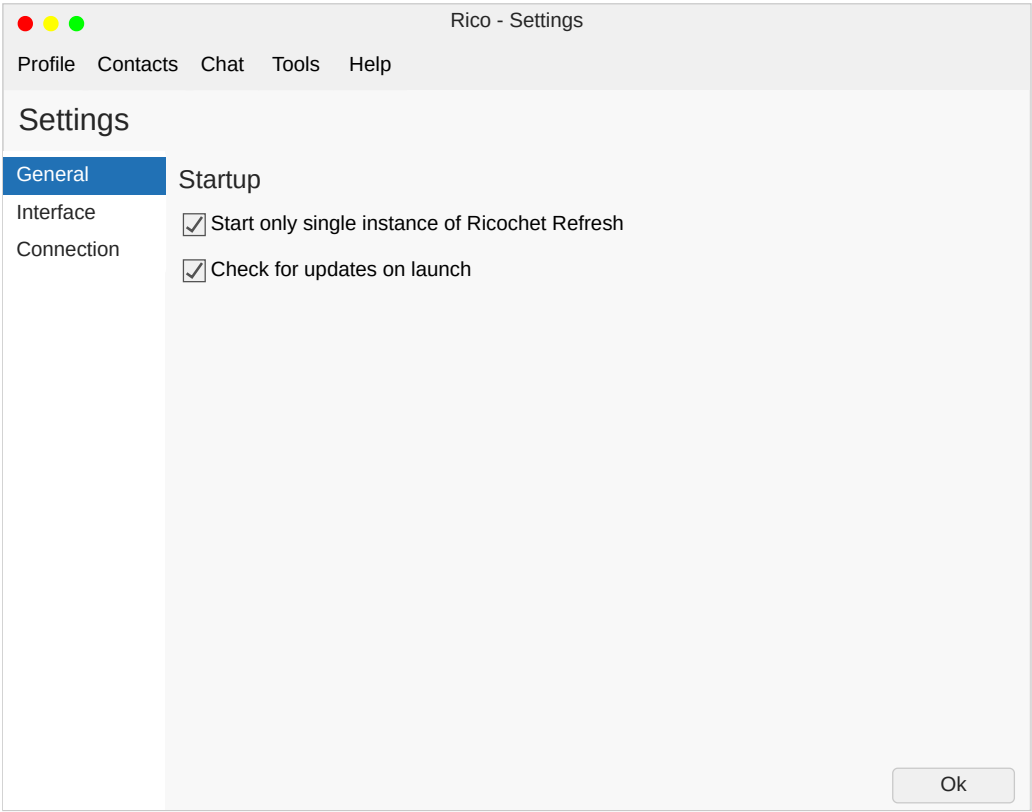
Copy Logs to Clipboard

Close

Rico - Settings (General)

The General section of the Preferences pane. Should enable configuring general application behaviour that are not specific to the other sections.

All options in this dialog must be safe to store un-encrypted in the application config; no privacy or security related settings belong here! Those must be saved in the profile itself.



Rico - Settings (Interface)

The Interface section provides users with UI customisation options

Rico - Settings

ProfileContactsChatToolsHelp

Settings

General

Interface

Connection

Language

Select interface language:

English (en-US)

Alerts

☒

Show desktop notifications

☒

Blink taskbar icon

☒

Enable audio notifications

Window

☒

Minimize instead of exit

☒

Show system tray icon

☒

Minimize to system tray

Ok

Languages sorted by language code

Rico - Settings (Connection)

The Connection section provides users with Tor Provider related options. This page may vary a lot depending on which Tor Provider is selected. For instance, a System Tor Daemon Tor Provider (eg on Tails) would have this section almost entirely empty. For now let's have this mostly mirror about:preferences#connection in Tor Browser.

This entire section will be disabled if the user is already connected to the Tor Network

Rico - Settings

ProfileContactsChatToolsHelp

Settings

General

Interface

Connection

Quickstart

Quickstart connects Rico to the Tor Network automatically when launched, based on your last used connection settings

☒ Always connect automatically

Backend

Select which Tor client implementation to use. Changing requires an application restart.

☒ Out-of-Process Tor daemon (default)

☐ System Tor daemon

☐ In-Process Arti

Bridges

Bridges help you securely access the Tor Network in places where Tor is blocked. Depending on where you are, one bridge may work better than another.

☒ Use Bridges

Add bridges

☒ Chose from one of Ricochet Refresh's built-in bridges

☒ obfs4

Makes your Tor traffic look like random data. May not work in heavily censored regions.

☐ Snowflake

Routes your connection through Snowflake proxies to make it look like you're placing a video call, for example.

☐ meek-azure

Makes it look like you're connected to a Microsoft website, instead of using Tor. May work in heavily censored regions, but is usually very slow.

☐ Enter bridge addresses you already know

Paste your bridge addresses here

Network Settings

☐ Use Proxy

Proxy Type

SOCKS5

Address

IP address or hostname

Port

Username

Optional

Password

Optional

☐ Configure Firewall

Allowed Ports

80,443

Ok

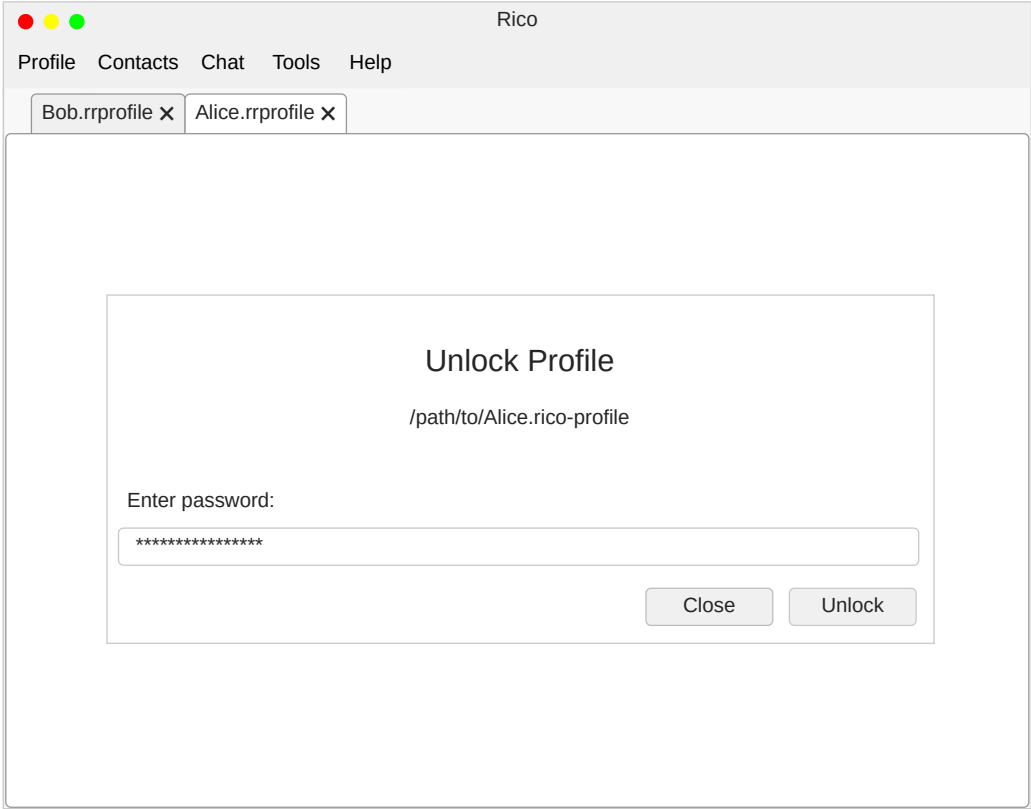
Add bridges region collapses if Use bridges is not ticked

Rico - Unlock Profile

Users reach this window after connecting to the Tor Network and opening a profile. A password is required to decrypt a Rico profile. Obviously we draw a lot of inspiration here from KeyPassXC and friends.

Menus

- | | | | | | |
|----------------------|----------------------|------------------|----------------|------------------------|------------------------------|
| Profile: | Contacts: | Chat: | Tools: | Help: | |
| - New Profile... | - Add Contact... | - Export Logs... | - Downloads... | - Manual | - Licenses menu option |
| - Import Profile... | - Delete Contact | - Delete Logs... | - Tor Logs... | - Changelog... | should open a folder |
| - Open Profile... | - Connect/Disconnect | | - Settings... | - Licenses... | containing relevant licenses |
| - Save Profile As... | Contact | | | --- | - Changelog menu option |
| - Edit Profile... | - Block/Unblock | | | - Check for Updates... | opens CHANGELOG.txt in |
| --- | Contact | | | - About | system default text editor |
| - Close Profile | | | | | |
| - Logout | | | | | |
| --- | | | | | |
| - Copy User Id | | | | | |
| - Set Visibility | | | | | |
| - Visible | | | | | |
| - Restricted | | | | | |
| - Hidden | | | | | |
| - Offline | | | | | |
| --- | | | | | |
| - Logout All | | | | | |
| - Quit | | | | | |



Users will always 'login' into the 'offline' state, they will opt-in to less private visibility states

Rico - New Profile

Accessed via New Profile... menu item in File menu. Walks the user through creating a new Rico profile.

Label

Start

Rico

ProfileContactsChatToolsHelp

Create New Profile

Step 1 of 3

New profile destination:

/home/alice/alice.rr-profile

Browse...

Back

Cancel

Next

Credentials

Rico

ProfileContactsChatToolsHelp

Create New Profile

Step 2 of 3

Display name:

Alice

Password:

Confirm password:

Back

Cancel

Next

Summary

Rico

ProfileContactsChatToolsHelp

Create New Profile

Step 3 of 3

New profile destination:

/home/alice/alice.rr-profile

Ricochet Refresh id:

ricochet-v4://t3dt6eazvm5xu3fxaumhdg7jjgdnj4bws...

Back

Cancel

Finish

Button copies user's ricochet id to clipboard

Finish writes new profile to disk and focuses new profile tab in the main chat view

Rico - Import Profile

Accessed via Import Profile... menu item in File menu. Walks the user through import a legacy Rico profile.

Label

Start

Rico

ProfileContactsChatToolsHelp

Import Legacy Profile

Step 1 of 3

Legacy profile location:

/home/alice/.config/ricochet-refresh/ricochet.js...

Browse...

New profile destination:

/home/alice/alice.rr-profile

Browse...

Back

Cancel

Next

Credentials

Rico

ProfileContactsChatToolsHelp

Import Legacy Profile

Step 2 of 3

Display name:

Alice

Password:

Confirm password:

Back

Cancel

Next

Summary

Rico

ProfileContactsChatToolsHelp

Import Legacy Profile

Step 3 of 3

New profile destination:

/home/alice/alice.rr-profile

Ricochet Refresh id:

ricochet-v4://t3dt6eazvm5xu3fxaumhdg7jjgdnj4bws...

Back

Cancel

Finish

Button copies user's ricochet id to clipboard

Finish writes new profile to disk and focuses new profile tab in the main chat view

Rico - Profile

Profile view/edit is necessarily part of a loaded profile, so it appears within a profile tab. The user can get the rico id associated with this profile here.

Rico

Profile

Contacts

Chat

Tools

Help

Bob.rrprofile ✕

Alice.rrprofile ✕

Profile

Profile destination:

/path/to/profile.rico-profile

Rico ID:

rico:6l62fw7tqctlu5fesdqukvpoxezkaxbzllrafa2ve6ewuhzphxczsjyd

Display name (required):

Name shown to your contacts

Pronouns (optional):

Your preferred pronouns

Avatar (optional):

Path to avatar image

Browse...

Image will be rescaled to 256 x 256 pixels

Remove

Description (optional):

Text description displayed in your user profile

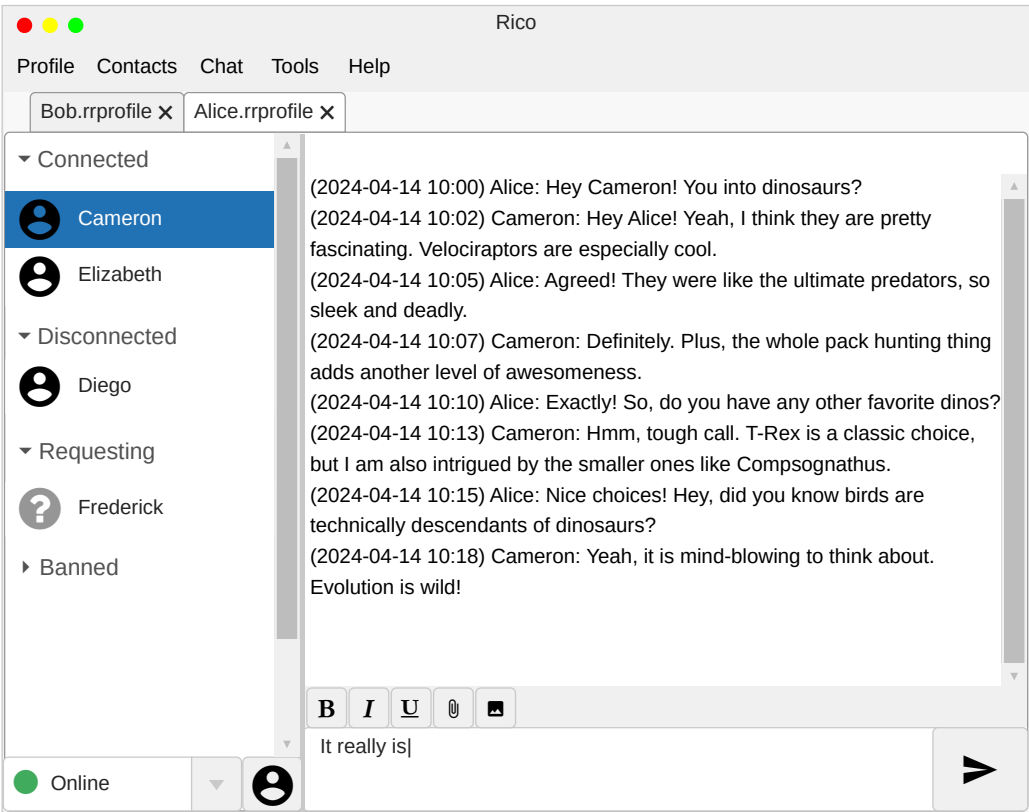
1024 characters available

Cancel

Ok

Rico - Conversations (Online)

Profile tab when the user is logged in. In 'Online' mode identity and endpoint servers are enabled, anyone can cyberstalk the user if they know the user's Identity Server Service ID



- Hide a conversation by clicking on a user entry again to unselect

- Per-contact context menu will have all the per-contact commands from the Contacts menu

User groups:

- Connected: users which have connected to our, or which we have connected to their endpoint server

- Connecting: users which we are explicitly attempting to connect to

- Disconnected: no active connection; offline to us

- Requesting: in-progress outgoing friend requests

- Banned: no endpoint server, actively blocking attempts when accessing our identity server

Visibility Modes:

- Online: identity and endpoint servers are enabled

- Restricted: endpoint servers are enabled

- Hidden: no endpoints enabled, but outgoing connections can occur

- Offline: no endpoints enabled, no outgoing connections either

NOTE: Online should really communicate most risk, users can be cyberstalked if they always stay in Online mode

Minor text formatting:

- Bold
- Italic

- Underline

Add Attachment

Inline-image

Profile button:

Open an Edit Profile dialog where user can:

- get Ricochet id
- set display name
- set profile image
- set profile info
- w/e else metadata users opt into

Minor text formatting:

- Bold
 - Italic
 - Underline
- Add Attachment
- Inline-image

NOTE:

- Attachments are sent as-is, but are not rendered/previewed by the application.
- Inline-images will be somehow re-encoded to hopefully avoid malformed image decoder bugs/exploits

- URLs will never open a browser, only open a context menu with the option to copy the URL to clipboard - exported conversations should be standardised to avoid leaking user's locale re timestamps, any potential formatting options, etc - user status (connect/disconnect) messages will appear in chat

TODO: Chat Thumbnail and Image UI; images should have be optionally spoiler'd and have text descriptions

TODO: Incoming Friend Request flow

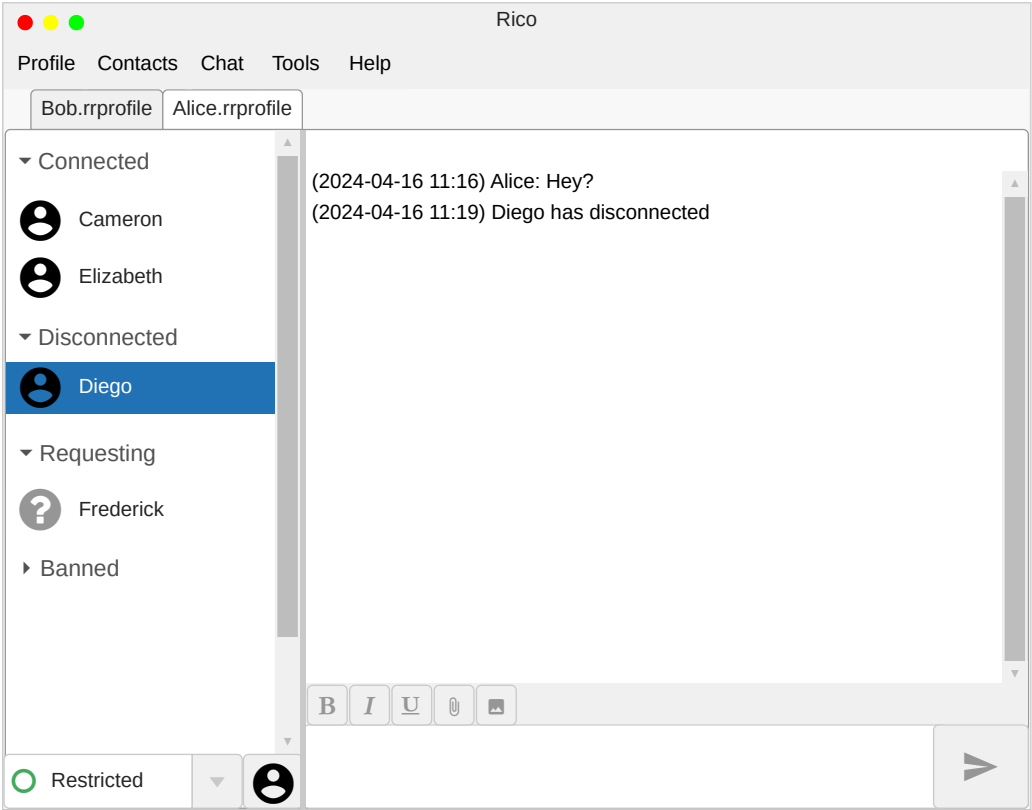
TODO: Status bar with: Onion Service status, reachability, service id

TODO: Smily selector

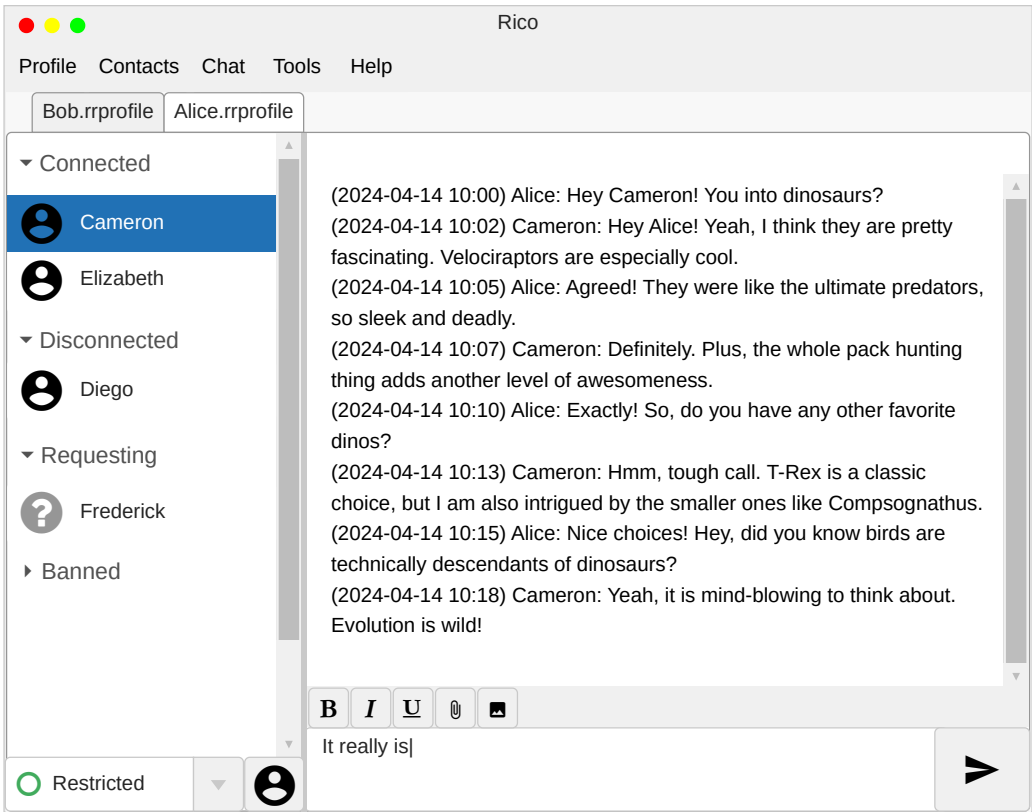
TODO: Invite-only connectivity to identity server using client auth

Rico - Conversations (Restricted)

In 'Restricted' mode (identity server disabled, endpoint servers enabled) only confirmed/allowed contacts may contact us. New contacts will not be able to access the public identity server to get through the gosling handshake to get their own endpoint server. This way strangers will not be able to cyber-stalk a user based off of their (potentially) public identity server. The UX should be identical to Online mode, except the user will never receive new friend requests.

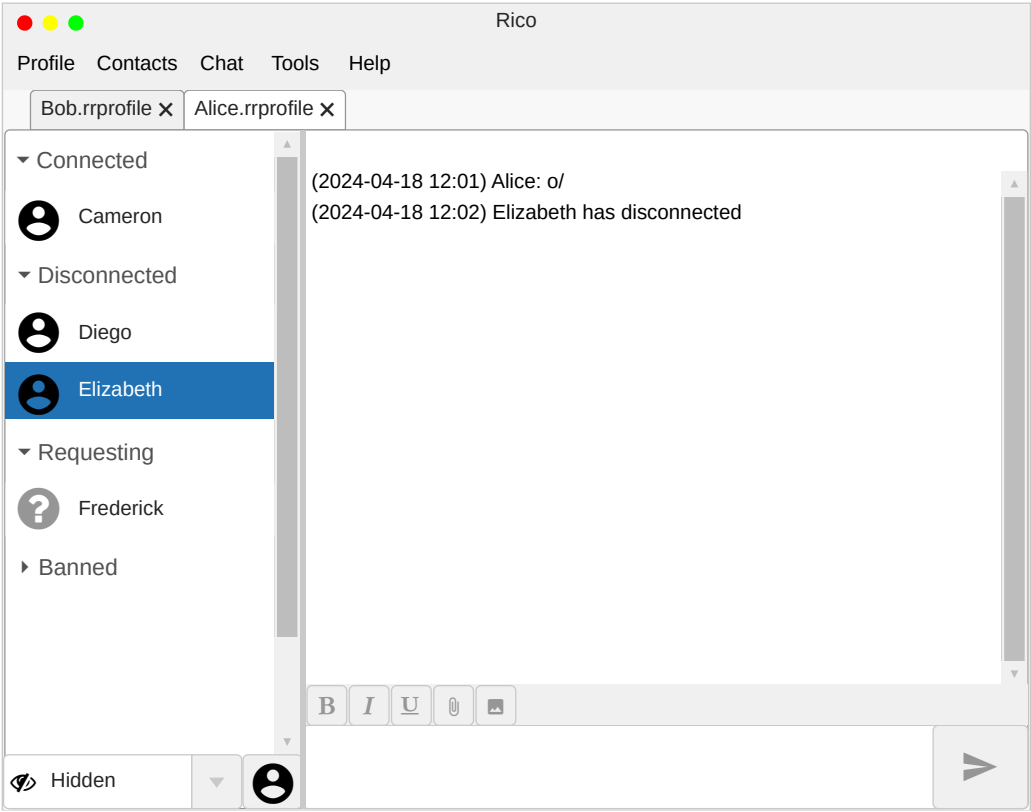


- Chat input widgets are disabled if the user is not connected



Rico - Conversations (Hidden)

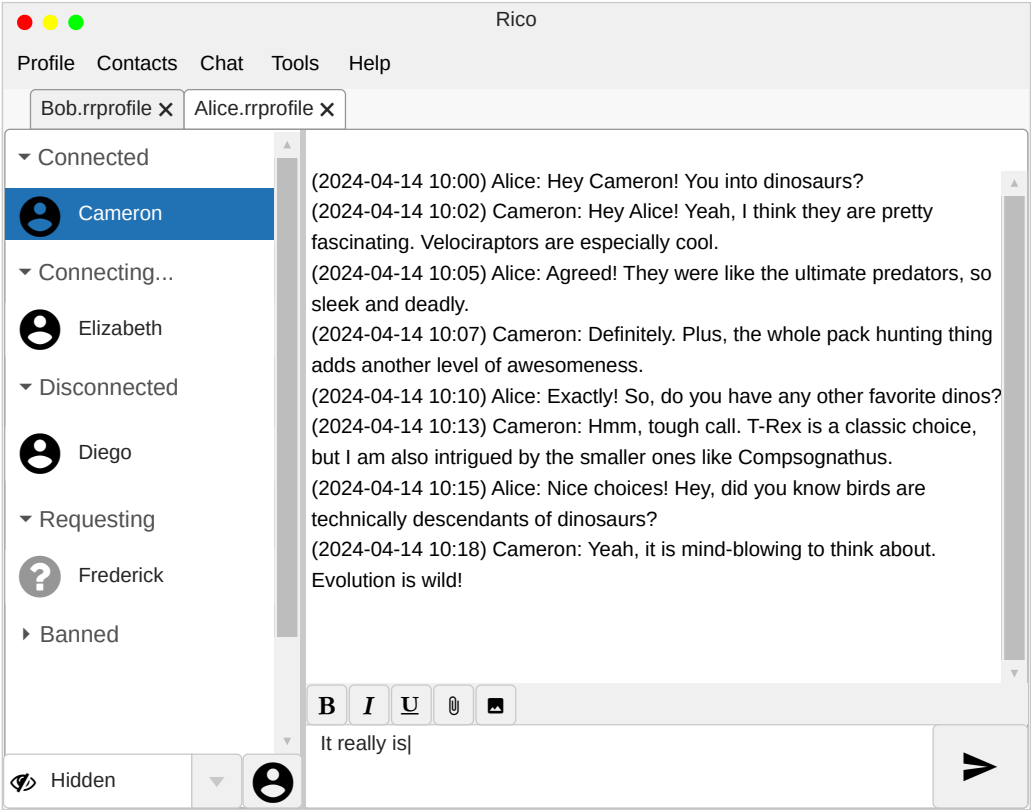
In 'Hidden' mode we do not enable any of our endpoint servers or identity servers, so even contacts do not gain our online/offline metadata. Connections are outgoing only and must be explicitly initiated by the user. This does imply that two users which are both in 'Hidden' mode will not be able to ever connect to each other!



- Chat input widgets are disabled if the user is not connected

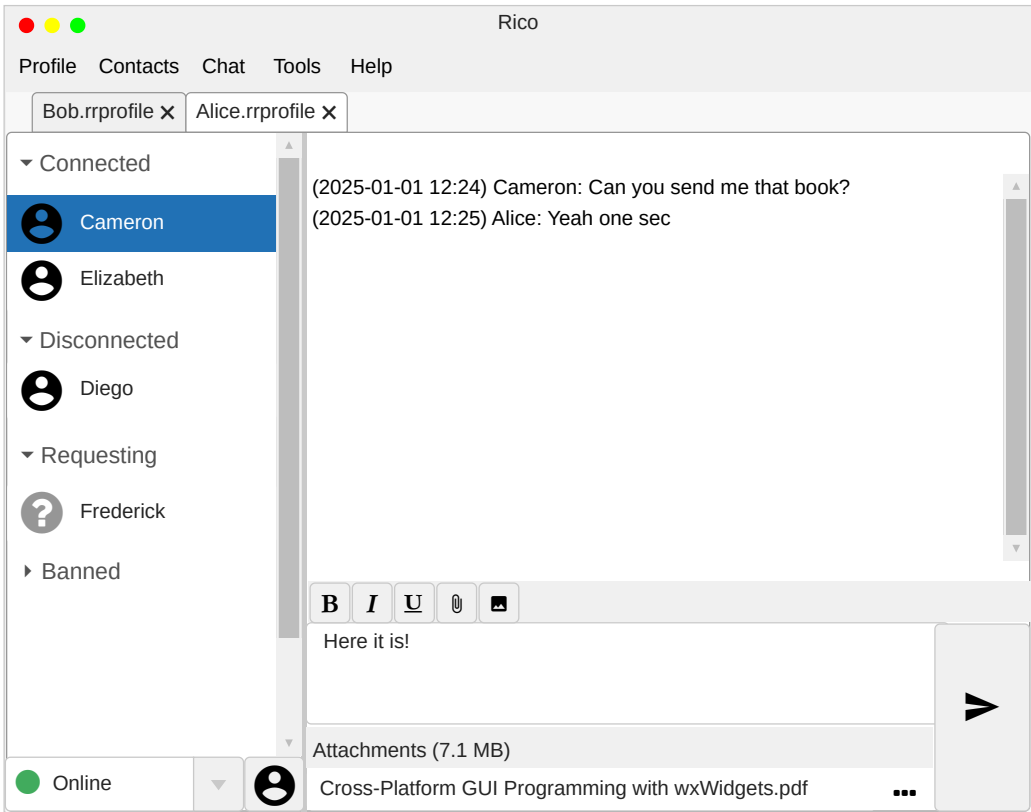
- User must opt-in to connecting to contacts when in Hidden mode using the chat icon in the toolbar or connect action in the contacts menu

- Contact goes into Connecting... group when connect is initiated
- Once a user reaches Connected, then chat can happen



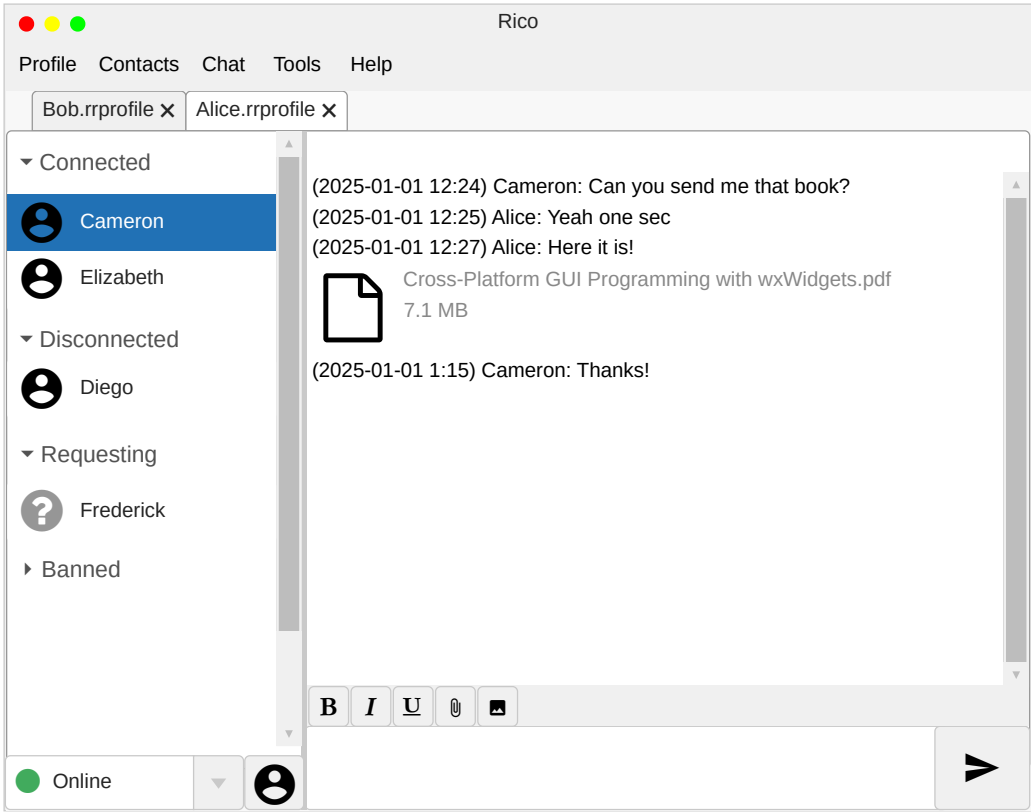
Rico - Attachments

Attachments in general appear in-line with chat messages.



Attachment button opens file selection dialog; selected file added to Attachments section. Attachment message is sent like an ordinary message

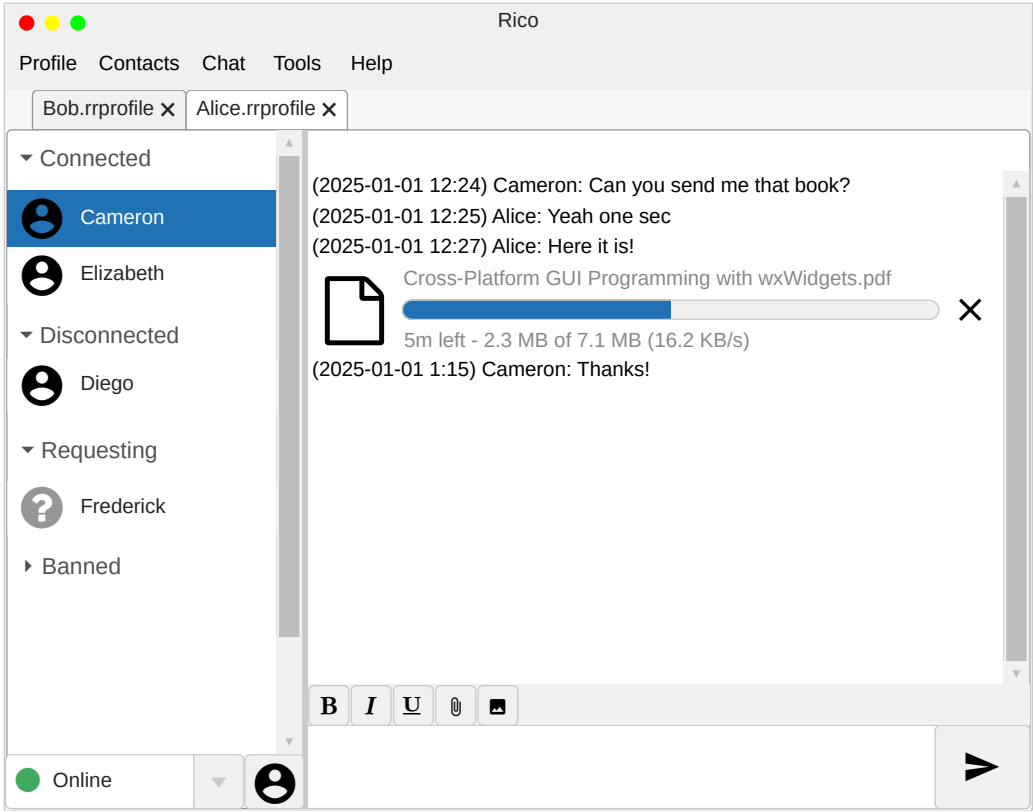
Per-attachment Meatball menu:
- Remove
- Open Folder



User initiates download by clicking on the attachment preview widget

Rico - Attachments (continued)

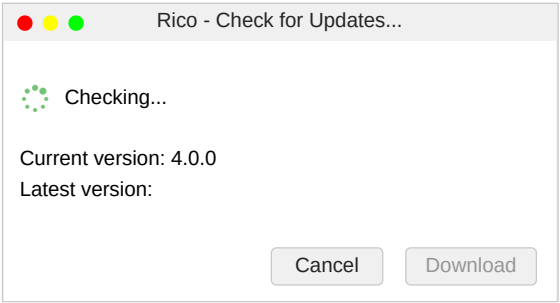
Attachments in general appear in-line with chat messages.



User initiates download by clicking on the attachment preview widget; on cancel widgets goes back to original ready to download state

Updates Dialog

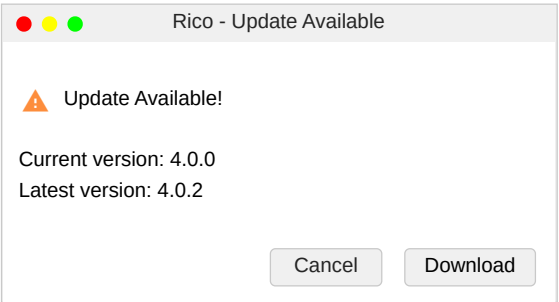
Accessed via the Help > Check for Updates... menu item. This will open a dialog which pings an onion service for update jsons, which will tell this dialog which install media/package to download.



- Check happens over Tor to Onion Service endpoint
- Will need some to maintain some update jsons



- Ricochet Refresh is up to date so we good



- Download the correct package based on the current insalled version (eg a deb, appimage, dmg, windows installer, etc

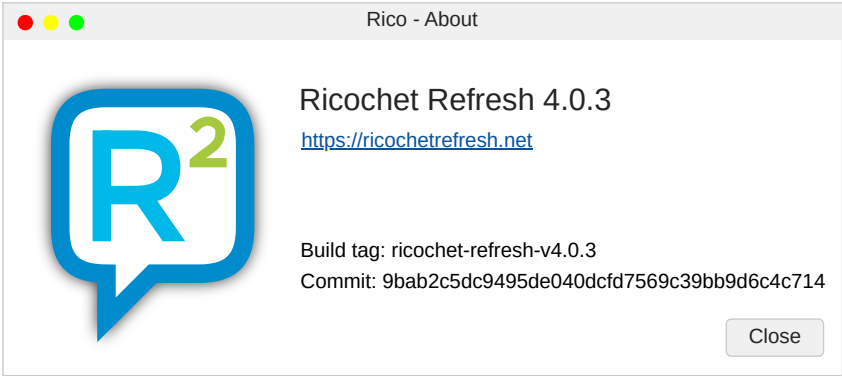
- Download button will open Downloads dialog

About Dialog

Accessed via the Help > About menu item. Dialog should show simple version and build info

Stable Channel

Like in chats, the URL will open a context menu allowing the user to copy the link to the clipboard



Ensure tag and commit labels are copy-able

Alpha Channel

